

# Khushal Midha

in LinkedIn: Khushal Midha </> CodeForces </>  
GitHub: khushalmidha </> CodeChef </> LeetCode

+91 90507 40836

midhakhushal5@gmail.com

## EDUCATION

Indian Institute of Information Technology, Lucknow  
B.Tech in Computer Science and Artificial Intelligence – CGPA: 8.72/10

Courses: OOPS, DBMS, OS, DSA  
August 2023 – July 2027

## EXPERIENCE

SDE Intern – Coffeee.io Remote — Jun 2025 – Aug 2025

- Built a **project-based assessment module** (Java, Spring Boot, React, PostgreSQL) where candidates debug live codebases via **error logs**, graded automatically against hidden test cases.
- Developed a **read-only AI assistant** (LLMs, Redis, Docker) to provide contextual hints without edit access, preserving evaluation integrity; validated in **staging**.

## PROJECTS

- 1. MediPulse** Feb 2025 – Present  
MERN, FastAPI, Hugging Face Transformers, Redis, Kafka, Socket.IO, WebRTC, RAG Live Demo  
  - Built a **multi-tenant healthcare SaaS** with **125 REST APIs**, **24 MongoDB models**, and **34 React pages**, supporting **appointments**, **live OPD queues**, **WebRTC consultations**, and **hospital portals** with **7-role RBAC** via **JWT & Google OAuth**.
  - Developed an **AI triage engine** combining a **Gemini agent**, **3 Hugging Face models**, and a custom **RAG pipeline** (using **text-embedding-004 & Cosine Similarity**) via **FastAPI** for accurate **disease prediction**, **specialty routing**, and **context-aware patient memory**.
  - Engineered **real-time OPD queues** and **virtual wallets** using **Socket.IO**, **Redis**, and **Kafka**, implementing **atomic transactions**, **distributed locks**, **idempotent refunds**, and **asynchronous event processing** for **concurrency-safe operations**.
- 2. QuantDesk** Jan 2026 – Present  
C++, Python, React, Quantitative Trading, CMake Live Demo  
  - Built a deterministic **C++ order book & matching engine** with **price-time priority**, partial fills, **LIMIT/MARKET/IOC/FOK orders**, and **event-driven execution** with **GoogleTest** coverage.
  - Implemented **Avellaneda-Stoikov market making** and **cointegration-based statistical arbitrage** in Python, with a **React dashboard** for **strategy replay** and **visualization**.
- 3. ThreatGraph** Jan 2026 - Present  
Python, PyTorch, FastAPI, Kafka, PostgreSQL, NetworkX, FAISS, RAG, GNNs  
  - Architected a **spatial-temporal threat detection engine** fusing **Graph Neural Networks** and **Transformers**, processing **3,000+ events/sec** via **Kafka**, achieving **92% PR-AUC** on imbalanced attacks.
  - Built a full-stack **automated incident response platform** integrating a **RAG-powered LLM analyst** to map attack paths to **MITRE ATT&CK** and trigger **real-time endpoint quarantines** via **FastAPI**.

## ACHIEVEMENTS

- Rated as **Expert** Codeforces(1809), **5-star** CodeChef (2137), **Guardian** LeetCode (2137).
- Solved **2500+ DSA problems** across Codeforces and LeetCode strengthening problem-solving skills.
- Selected for **Amazon ML School 2025**, among top 3000 from 85,000 applicants Proof
- ICPC Amritapuri Regionalist 2024-25**, securing a **rank of 57** with team **Overshadowed**.
- Secured rank of **186** in **Facebook Meta Hackercup Round-3** among 14000+ Participants Ranklist
- Topped among **250+ peers** in **Freshers Cup 2024** by CP Wing, Axios, IIIT Lucknow
- Achieved All India Rank **15,503** in **JEE Advanced 2023** and Rank **9,932** in **JEE Mains 2023**.

## SKILLS & OTHERS

- Languages:** C, C++, Python, JavaScript, Java, SQL
- Frontend:** React.js, TypeScript, HTML, CSS, Tailwind CSS, Bootstrap
- Backend:** Node.js, Express.js, Spring Boot, Redis, Kafka, Fast API
- Database:** MongoDB, MySQL, PostgreSQL
- Machine Learning & AI:** PyTorch, XGBoost, Scikit-learn, Transformers, MLOps (MLflow), LLMs, Agentic AI, Prompt Engineering
- Courses:** Object Oriented Programming, Database Management System, Computer Networks, Operating System
- Technical Skills:** Data Structures Algorithms, Competitive Programming, Web Development, REST API Design